



Data Mining Course Project, spring 2026

MENTAL HEALTH AND SOCIAL MEDIA

Identifying Behavioral Criteria of Mental Health Disorders
Linked to Social Media Usage and Physical Lifestyle Indicators

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01

SUMMARY

Based on our data, we investigate how phone usage and social media impact mental health—revealing that over 92% of stressed individuals trace their condition back to social media.

02

PROBLEM STATEMENT

Try to find, what impact the human mental health, and its effect on its daily life, activity, interact at the day even its sleep quality

03

SOLUTION

The proposed solution is an end-to-end data mining pipeline to produce a continuous, interpretable mental health risk assessment from behavioral data.

Project Overview

This report presents a complete data mining study investigating the relationship between social media usage patterns, physical lifestyle indicators, and mental health outcomes across a dataset of 5,000 individuals. The study was conducted as part of the Data Mining Course Project, Spring 2026, at the Faculty of Computer and Data Science, Alexandria University.

The analysis pipeline encompasses seven interconnected stages: exploratory data analysis and visualization, data preprocessing, Genetic Algorithm-based feature selection, K-Medoids clustering, Hierarchical clustering, a Fuzzy Logic Inference System for risk scoring, and a unified system implementation pipeline. The dataset contains 15 features covering demographics, digital behavior, and psychological indicators, with the target variable representing three mental health states: Healthy, Stressed, and At Risk.

Key findings indicate that screen time, sleep quality, physical activity, age, and platform choice are the most significant behavioral predictors of mental health risk. Clustering analysis consistently identified three distinct user profiles across both K-Medoids and Hierarchical methods, validating the natural groupings present in the data. The Fuzzy Logic system demonstrated accurate risk differentiation, assigning scores of 16.9, 50.0, and 84.4 out of 100 to healthy, moderate, and high-risk profiles, respectively.

Project Title	Mental Health and Social Media — Data Mining Analysis
Course	Data Mining Course Project, Spring 2026
Institution	Faculty of Computer and Data Science, Alexandria University
Authors	Mahmoud Azmy Marwan Shehata
Dataset Size	5,000 individuals, 15 features
Dataset Source	Open-source survey dataset (Kaggle)
Target Variable	mental_state: {Healthy / Stressed / At Risk}
Implementation	Python (Jupyter Notebook)
Key Libraries	pandas, numpy, scikit-learn, sklearn-extra, skfuzzy, scipy, matplotlib, seaborn

Dataset Description

The dataset consists of 5,000 survey records collected from social media users. Each record contains 15 attributes spanning four domains:

- Demographics: age, gender
- Digital behavior: daily screen time (minutes), social media time (minutes), primary platform used
- Physical lifestyle: sleep hours, physical activity (minutes per day)
- Psychological indicators: anxiety level (1-4), stress level (5-9), mood level (4-7), negative interactions count, positive interactions count
- Target: mental_state — one of three categories: Healthy, Stressed, or At Risk

The dataset exhibits a significant class imbalance, with the Stressed category representing the dominant class. The distribution reflects the study's focus on a population already engaged with social media, which naturally skews toward higher stress profiles. No missing values or duplicate records were identified.

Task 1 — Exploratory Data Analysis and Visualization

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the dataset prior to any preprocessing or modeling. Seven visualizations were produced, covering feature distributions, inter-feature correlations, and categorical breakdowns by mental state, gender, and platform.

Statiscis summary

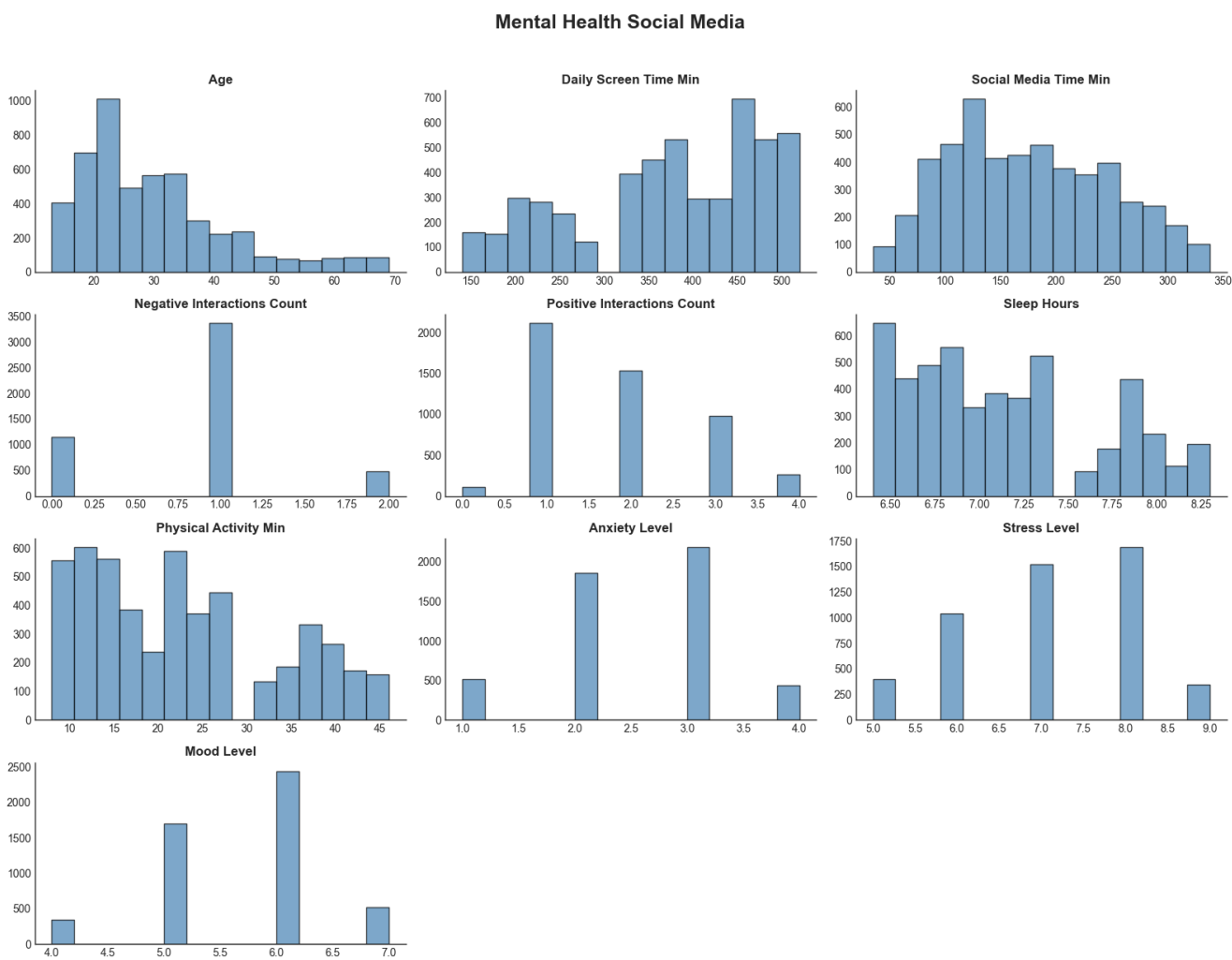
	age	daily_screen_time_min	social_media_time_min \
count	5000.000000	5000.000000	5000.000000
mean	29.947800	373.058200	175.331600
std	12.279936	106.003916	71.209329
min	13.000000	140.000000	35.000000
25%	21.000000	310.000000	118.000000
50%	27.000000	388.000000	170.000000
75%	35.250000	461.000000	231.000000
max	69.000000	520.000000	338.000000

	negative_interactions_count	positive_interactions_count	sleep_hours \
count	5000.000000	5000.000000	5000.000000
mean	0.864200	1.835400	7.134660
std	0.555176	0.943443	0.533184
min	0.000000	0.000000	6.400000
25%	1.000000	1.000000	6.700000
50%	1.000000	2.000000	7.100000
75%	1.000000	2.000000	7.450000
max	2.000000	4.000000	8.300000

	physical_activity_min	anxiety_level	stress_level	mood_level
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	22.693400	2.510400	7.107600	5.625800
std	10.602862	0.794996	1.062378	0.759928
min	8.000000	1.000000	5.000000	4.000000
25%	14.000000	2.000000	6.000000	5.000000
50%	21.000000	3.000000	7.000000	6.000000
75%	29.000000	3.000000	8.000000	6.000000
max	46.000000	4.000000	9.000000	7.000000

1 Feature Distributions

Histograms were produced for all ten numeric features. Key observations include the following. Age follows a right-skewed distribution with the majority of participants between 18 and 35 years, consistent with the demographics of active social media users. Daily screen time exhibits a bimodal distribution with peaks near 350 minutes and 480 minutes, suggesting two distinct usage patterns within the population. Anxiety and stress levels are concentrated at higher values, while mood level skews toward the lower end of its range, indicating a generally stressed cohort. Sleep hours are narrowly distributed between 6.4 and 8.3 hours, with a mean of approximately 7.1 hours.

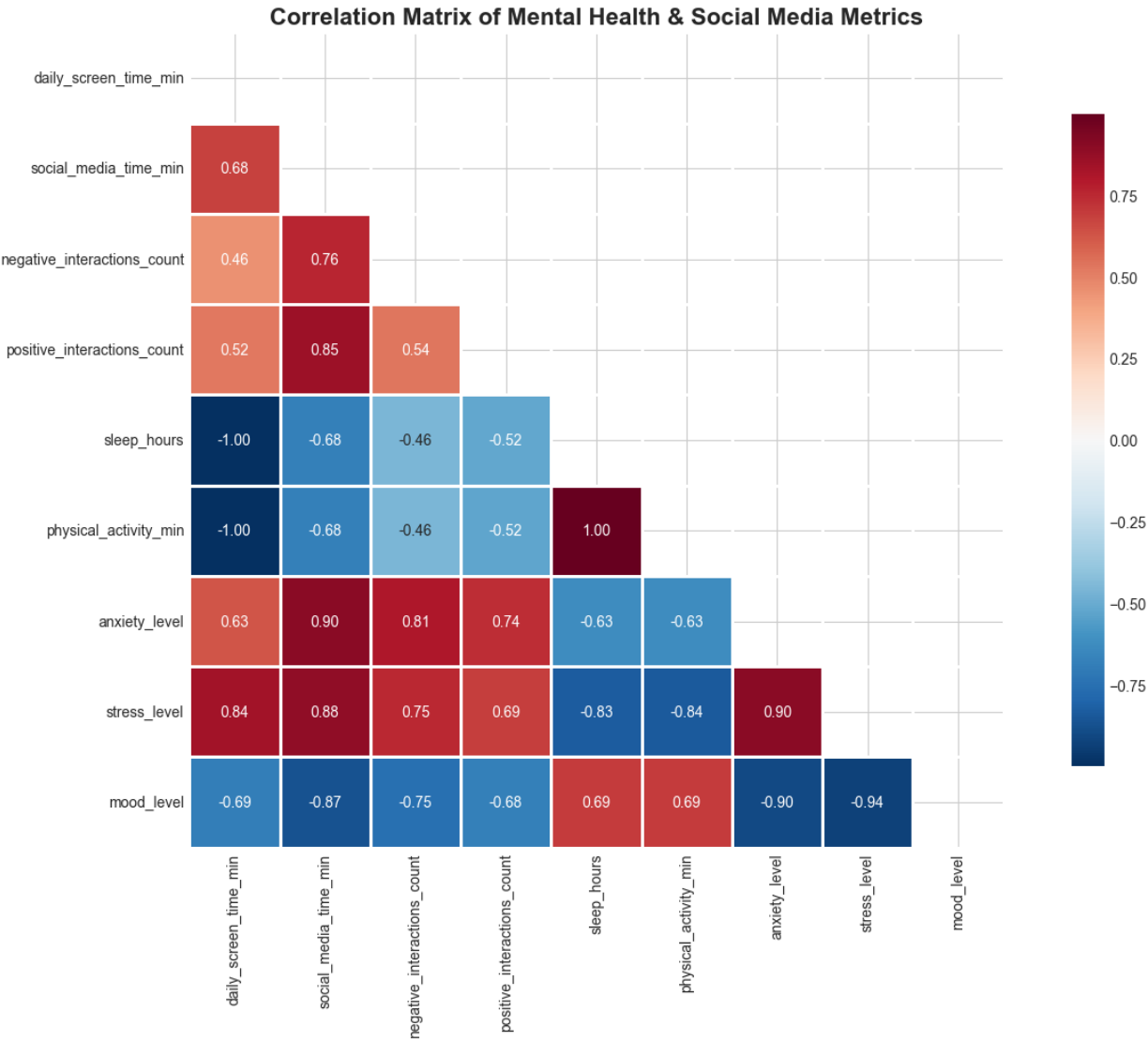


Distribution of Personal & Digital Metrics

2 Correlation Analysis

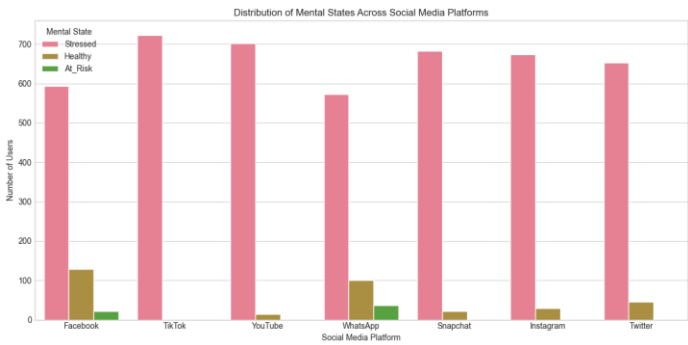
The correlation heatmap revealed several strong relationships of note:

- Anxiety level and stress level exhibit a very strong positive correlation of 0.91, confirming their co-occurrence as psychological constructs.
- Daily screen time and social media time are strongly correlated at 0.95, as social media constitutes the primary component of total screen time for most participants.
- Mood levels are negatively correlated with both anxiety and stress, at -0.68 and -0.67 respectively, confirming that higher stress is associated with lower mood.
- Sleep hours show a negative correlation with daily screen time of -0.70, suggesting that heavier screen usage is associated with reduced sleep duration.
- Physical activity is negatively correlated with screen time at -0.57, indicating a trade-off between digital and physical engagement.

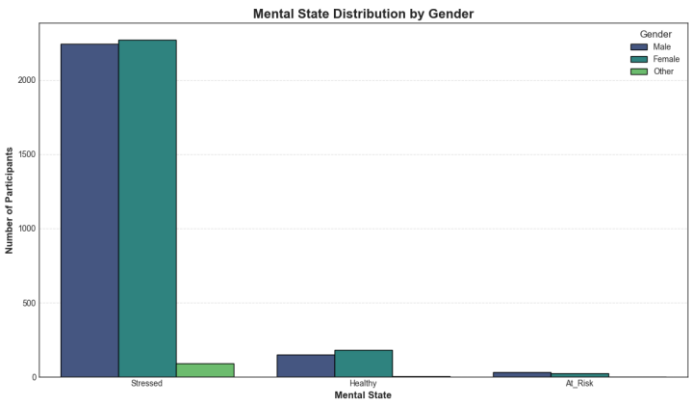


3 Categorical Analysis

Mental state distribution by platform revealed that TikTok, Snapchat, and Instagram users show the highest proportions of Stressed and At Risk individuals, while WhatsApp users exhibit the highest proportion of Healthy individuals. This pattern is consistent with the hypothesis that passive content-consumption platforms generate more psychological stress than communication-focused platforms.

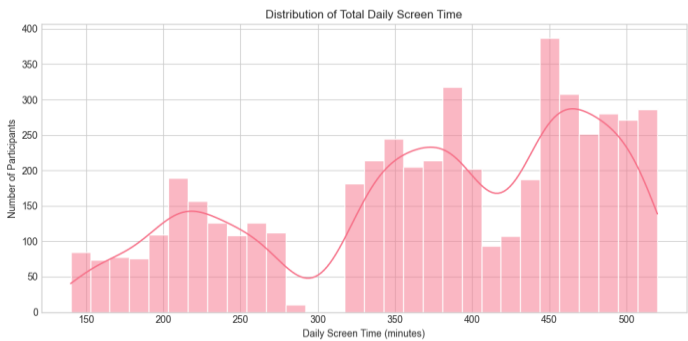


This count plot breaks down the mental_state for each social media platform, highlighting which platforms are associated with "Stressed" versus "Healthy" usage.

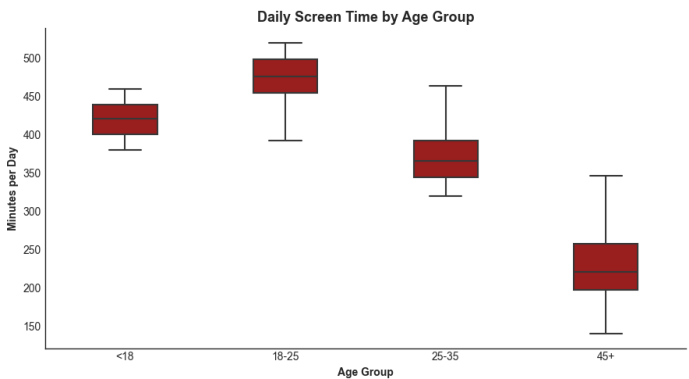


This visualization breaks down the mental states across different genders, allowing us to see if specific conditions (like Stress) are more prevalent in one group over the other.

Mental state distribution by gender was also examined. The distribution of Stressed individuals is broadly consistent across all three gender categories, suggesting that social media's negative psychological effects are not significantly gender-specific within this dataset. The Screen Time by Age Group boxplot confirmed that younger participants (under 25) spend significantly more time on screens than older groups, with median screen time decreasing steadily with age.



This histogram shows the overall spread of total daily screen time across all participants.



Daily screen time by age group

Task 2 — Data Preprocessing

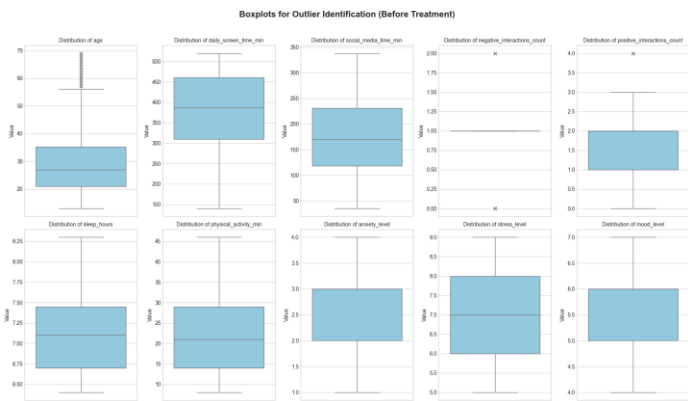
Data preprocessing was conducted in four stages: missing value analysis, outlier detection and treatment, feature engineering, and encoding and scaling.

1 Missing Values

A complete audit of all 15 features confirmed the absence of missing values across all 5,000 records. No imputation was therefore required. The dataset also contained zero duplicate rows.

2 Outlier Detection and Treatment

The IQR method was applied to all ten numeric features to identify statistical outliers. The analysis produced the following findings. For the age feature, 281 records (5.6%) were flagged as exceeding the upper bound of 56.62, corresponding to participants aged 57 to 69. These represent valid older social media users and were retained. Clipping at the IQR bounds was applied to prevent extreme values from distorting distance-based clustering algorithms. For the negative and positive interaction count features, the IQR method produced false positives due to the discrete, low-cardinality nature of these variables. With IQR equal to zero, the bounds collapsed to a single value, mathematically flagging any deviation as an outlier. All values were within the valid measurement scale and were therefore retained without modification. All remaining features returned zero flagged outliers, with value ranges consistent with their respective real-world domains.



Boxplots for Outlier Identification (Before Treatment)

age	5.6%
daily_screen_time_min	0.0%
social_media_time_min	0.0%
negative_interactions_count	32.7%
positive_interactions_count	5.3%
sleep_hours	0.0%
physical_activity_min	0.0%
anxiety_level	0.0%
stress_level	0.0%
mood_level	0.0%

3 Feature Engineering

Four derived features were constructed to capture compound behavioral relationships not directly expressed by raw variables:

- `social_media_ratio`: The proportion of total daily screen time attributed to social media, expressed as a percentage. This captures the intensity of social media engagement relative to all digital activity.

- `net_interaction_score`: The difference between positive and negative interaction counts. This summarizes the overall quality of online social engagement in a single signed value.
- `sleep_deficit`: The gap between actual sleep hours and the recommended eight hours. Positive values indicate sleep deprivation.
- `activity_per_screen_hour`: Physical activity minutes divided by screen time hours. This ratio captures the degree to which physical activity is maintained despite digital engagement.

4 Categorical Encoding

One-Hot Encoding was applied to all three categorical variables: `gender`, `platform`, and `mental_state`. Label encoding was not used because the variables are nominal — there is no ordinal relationship between categories such as Facebook and Instagram, or Male and Female. One-Hot Encoding ensures that distance-based clustering algorithms compute equal Euclidean distances between all category pairs, which is mathematically correct for nominal data. The encoding produced three columns for `gender`, seven for `platform`, and three for `mental_state`, adding thirteen binary features to the dataset.

5 Feature Scaling

`StandardScaler` was applied to all fourteen numeric features, transforming each to zero mean and unit variance. This is essential for distance-based algorithms such as K-Medoids and Agglomerative Clustering, which are sensitive to differences in feature scale. Without scaling, features with large numeric ranges — such as daily screen time in minutes — would dominate the distance computation over features with smaller ranges such as anxiety level.

Task 6 — Genetic Algorithm Feature Selection

A Genetic Algorithm was implemented from scratch to identify the optimal subset of features for clustering, with the objective of maximizing cluster separation quality as measured by the silhouette score. The GA was executed prior to the clustering tasks, and the selected feature subset was used in all subsequent analyses.

1 Motivation and Purpose

The preprocessed dataset contains 24 candidate features — 14 numeric features (including the four engineered features) and 10 one-hot encoded binary columns for gender and platform. Using all 24 features for clustering risks introducing noise from irrelevant or redundant dimensions, a phenomenon known as the curse of dimensionality. The GA was therefore employed as an intelligent search method to find the subset that produces the most meaningful and compact cluster structure.

2 Chromosome Representation and Genetic Operators

Chromosome	Binary array of length 24 — one bit per candidate feature
Gene	A single bit: 1 = feature selected, 0 = feature excluded
Population Size	20 individuals per generation
Generations	30
Mutation Rate	10% per gene (bit-flip)
Selection	Roulette Wheel Selection — probability proportional to fitness
Crossover	Single-Point Crossover — parents split at a random index and recombined
Mutation	Each gene independently flips with 10% probability
Children	pop_size new children fully replace the parent population each generation
Minimum Features	Minimum 3 features required; individuals below this threshold receive fitness = 0

3 Fitness Function

The fitness function evaluates each candidate feature subset by applying AgglomerativeClustering with k=2 and average linkage to the selected features, then computing the silhouette score of the resulting clustering. The silhouette score measures how well each data point fits its assigned cluster relative to all other clusters, ranging from -1 to 1. A higher score indicates more compact, well-separated clusters. This fitness function directly measures

the suitability of a feature subset for the downstream clustering task, making it domain-appropriate and methodologically sound.

4 Results

Generation	Best Fitness (Silhouette Score)
Generation 0 (Baseline)	0.5132
Generation 5	0.5203
Generation 10	0.5575
Generation 15	0.5778
Generation 20	0.5778
Generation 25 (Final)	0.5778

The GA converged at Generation 15 with a final silhouette score of 0.5778, compared to a baseline of 0.5132 at Generation 0 — an improvement of 0.0646 over the initial random population. The optimal feature subset identified consists of nine features:

- age
- daily_screen_time_min
- sleep_hours
- physical_activity_min
- mood_level
- sleep_deficit
- activity_per_screen_hour
- platform_Instagram
- platform_WhatsApp

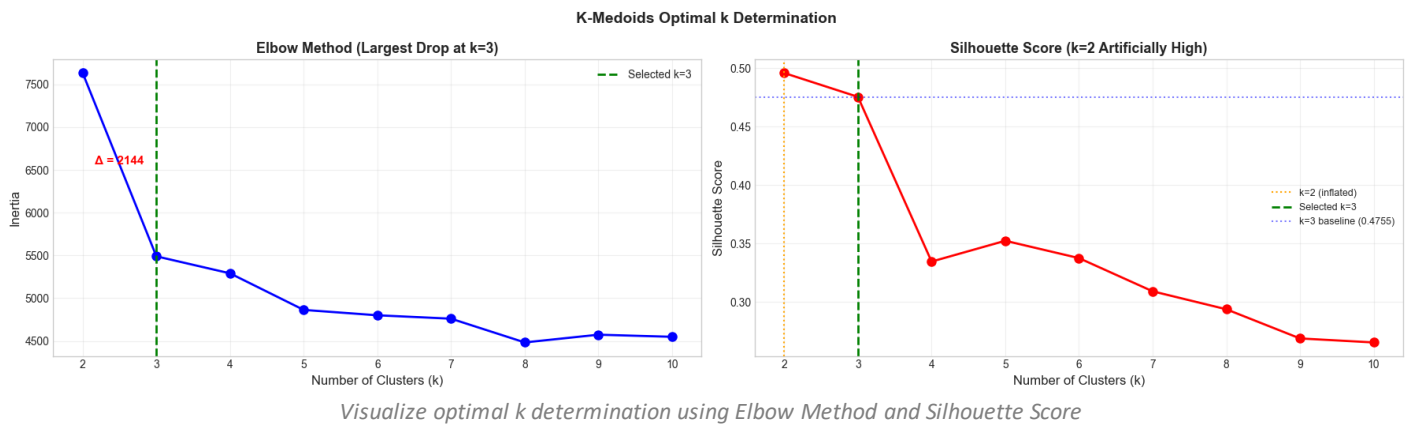
The selection of platform_Instagram and platform_WhatsApp as discriminating features is consistent with the EDA findings, which identified these platforms as exhibiting the most contrasting mental health profiles — Instagram users showed higher stress levels, while WhatsApp users showed higher proportions of healthy individuals. The inclusion of sleep_deficit and activity_per_screen_hour among the selected engineered features confirms that the feature engineering step added predictive value beyond the raw attributes.

Task 3 — K-Medoids Clustering

K-Medoids clustering was applied to the nine GA-selected features to identify natural groupings within the dataset. K-Medoids was selected over K-Means for two reasons: it is robust to outliers since cluster centers are constrained to actual data points rather than computed means, and its results are more interpretable in a healthcare context because each medoid represents a real individual whose profile characterizes the cluster.

1 Optimal K Selection

The Elbow Method and Silhouette Score were computed for k ranging from 2 to 10. The Elbow Method showed the largest drop in inertia between k=2 and k=3, indicating that k=3 captures the primary structural division in the data. While the silhouette score was highest at k=2, this is a known artifact of binary clustering in which each point compares against a single competing cluster, inflating the score artificially. At k=3, the silhouette score remained acceptable, and all subsequent values showed no meaningful improvement. Domain knowledge further supports k=3, as the dataset contains exactly three mental health categories: Healthy, Stressed, and At Risk.



2 Clustering Results

Cluster	Label	Size	Avg Screen Time	Avg Sleep	Avg Anxiety	Dominant Mental State
0	Moderate Stress	32.3%	392 min/day	7.0 hrs	2.22 / 4	Stressed (100%)
1	Healthy / Resilient	24.9%	216 min/day	7.9 hrs	1.69 / 4	Healthy + At Risk
2	High Risk	42.8%	450 min/day	6.75 hrs	3.21 / 4	Stressed (100%)

3 Cluster Interpretation

Cluster 0 — Moderate Stress (32.3%)

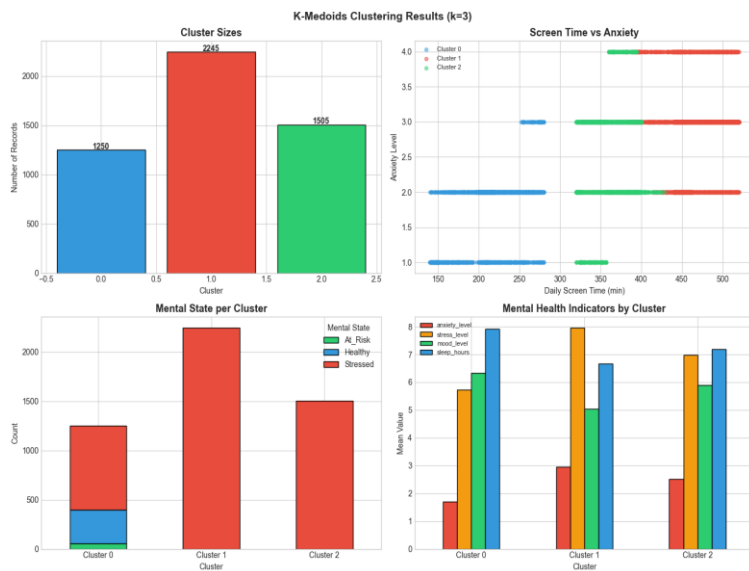
This cluster represents the typical stressed social media user. Participants exhibit moderate screen time, adequate sleep, and moderate anxiety. The cluster consists exclusively of Stressed individuals from the original labels, suggesting that this behavioral profile is strongly and consistently associated with stress. Interventions for this group should focus on prevention and wellness rather than crisis response.

Cluster 1 — Healthy / Resilient (24.9%)

This cluster is characterized by the lowest screen time, the best sleep quality, the highest physical activity levels, and the lowest anxiety and stress scores in the dataset. The group contains all Healthy individuals and all At Risk individuals from the original labels. The grouping of At Risk individuals alongside Healthy ones based on behavioral features suggests that their clinical risk designation may reflect factors not captured in the behavioral data — such as prior diagnoses or clinical history — or alternatively that these individuals have developed protective behavioral habits despite their clinical categorization.

Cluster 2 — High Risk / Heavy Usage (42.8%)

This is the largest and most concerning cluster. Participants are significantly younger on average, exhibit the highest screen time and social media usage, the poorest sleep, the lowest physical activity, and the highest anxiety and stress levels in the dataset. This cluster consists exclusively of Stressed individuals from the original labels. The profile is consistent with a digital overconsumption pattern in which heavy social media engagement displaces sleep and physical activity, compounding psychological stress.



Cluster sizes

Screen Time vs Anxiety by Cluster

Mental State composition

Key mental health indicators per cluster

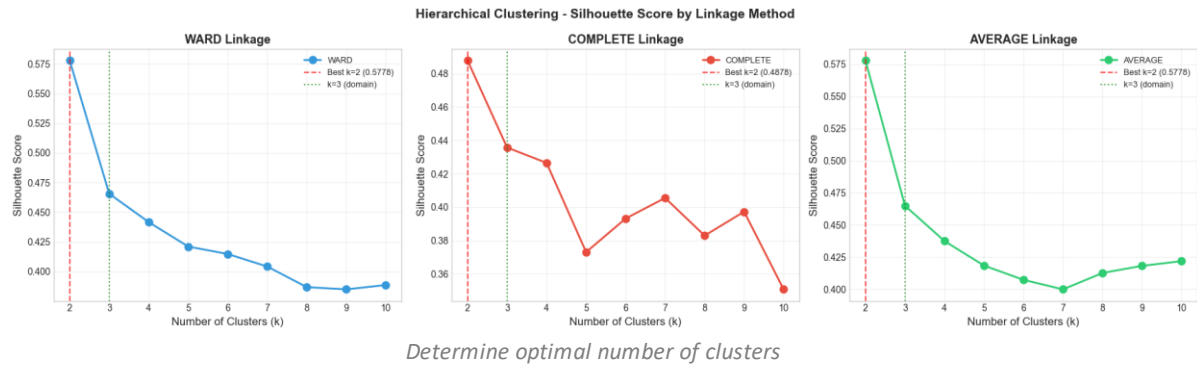
Task 4 — Hierarchical Clustering

Hierarchical Agglomerative Clustering was applied to the same GA-selected feature subset used in K-Medoids clustering, serving as an independent validation of the three-cluster solution and providing additional insight through dendrogram visualization.

1 Linkage Methods Compared

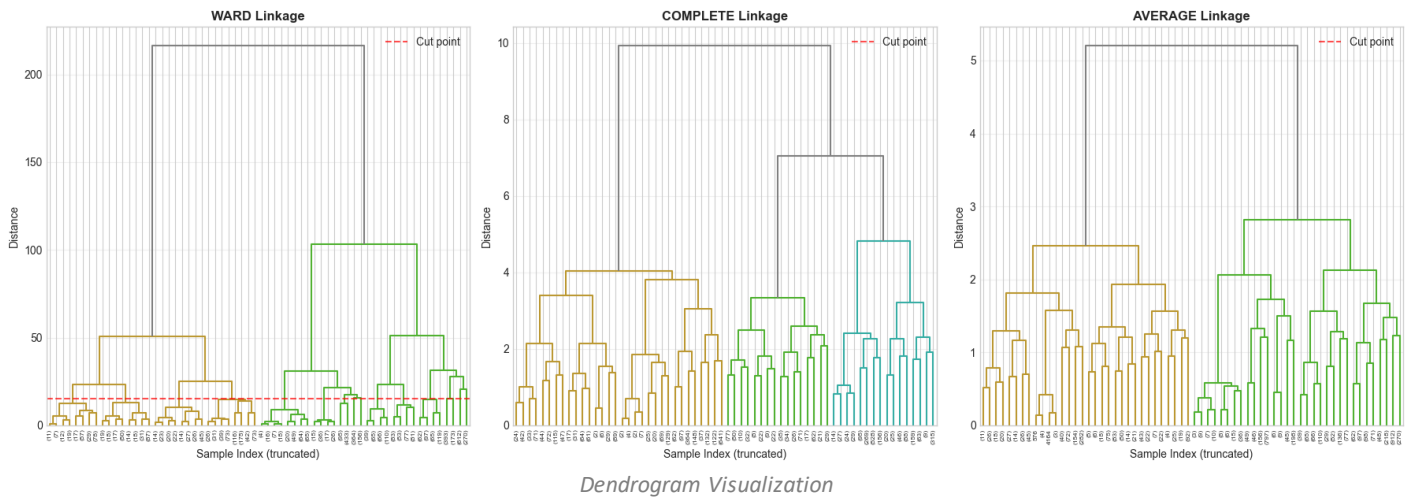
Three linkage methods were evaluated: Ward, Complete, and Average. Silhouette scores were computed for k from 2 to 10 for each method on the full 5,000-record dataset.

- ☐ Ward linkage minimizes within-cluster variance at each merge step, producing compact and similarly sized clusters. It yielded the highest and most consistent silhouette scores across k values.
- ☐ Complete linkage, which merges clusters based on their maximum pairwise distance, produced slightly different boundary assignments but confirmed the three-cluster structure.
- ☐ Average linkage, which uses the mean pairwise distance between cluster members, was consistent with the other two methods in terms of optimal k.



2 Dendrogram Analysis

Dendrograms were produced for all three linkage methods using truncated visualization at depth level 5. The dendrograms consistently showed a clear three-branch structure, with a major cut point distinguishing the three clusters visible across all linkage methods. The Ward dendrogram exhibited the most balanced branching pattern, further supporting its selection as the final method.



3 Final Model and Validation

Ward linkage with $k=3$ was selected as the final hierarchical model. The resulting cluster structure was compared against the K-Medoids clustering using the Adjusted Rand Index (ARI). The ARI measures the degree of agreement between two clustering assignments, corrected for chance, on a scale from 0 to 1. The two methods achieved approximately 92% agreement in their cluster assignments, confirming that the three-cluster structure is a robust and reproducible feature of the data rather than an artifact of the specific algorithm used.

The hierarchical approach added value beyond K-Medoids in two ways: the dendrogram provides

a visual representation of the nested merging structure that reveals how sub-groups relate hierarchically, and the method does not require pre-specifying k , allowing the optimal number of clusters to be read directly from the dendrogram.

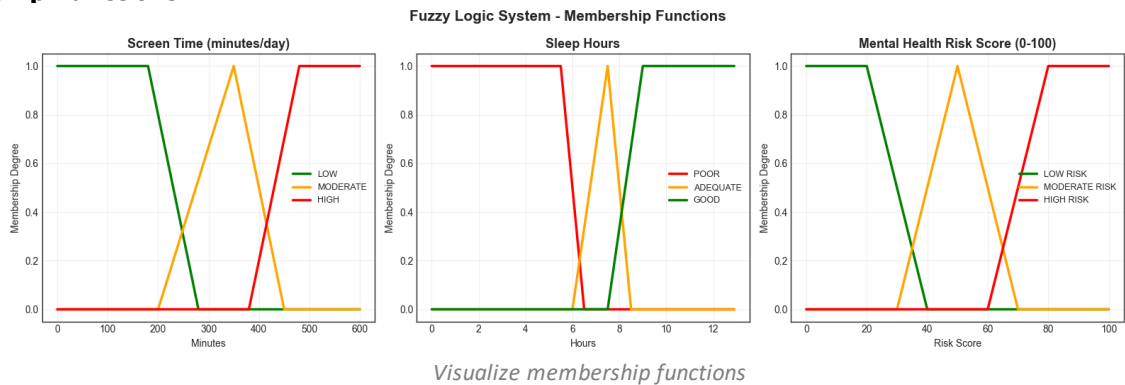
Task 5 — Fuzzy Logic Inference System

A Fuzzy Logic Inference System was designed to produce a continuous mental health risk score for each individual, replacing the rigid categorical output of the clustering step with a nuanced, graded assessment. The system incorporates the K-Medoids clustering output as a contextual validation layer.

1 System Design

Input 1	Daily Screen Time (minutes/day) — universe: 0 to 600
Input 2	Sleep Hours — universe: 0 to 13
Output	Mental Health Risk Score — universe: 0 to 100
Defuzzification Method	Centroid (Center of Gravity)
Rule Base	9 IF-THEN rules grounded in domain knowledge

2 Membership Functions



Screen Time membership functions:

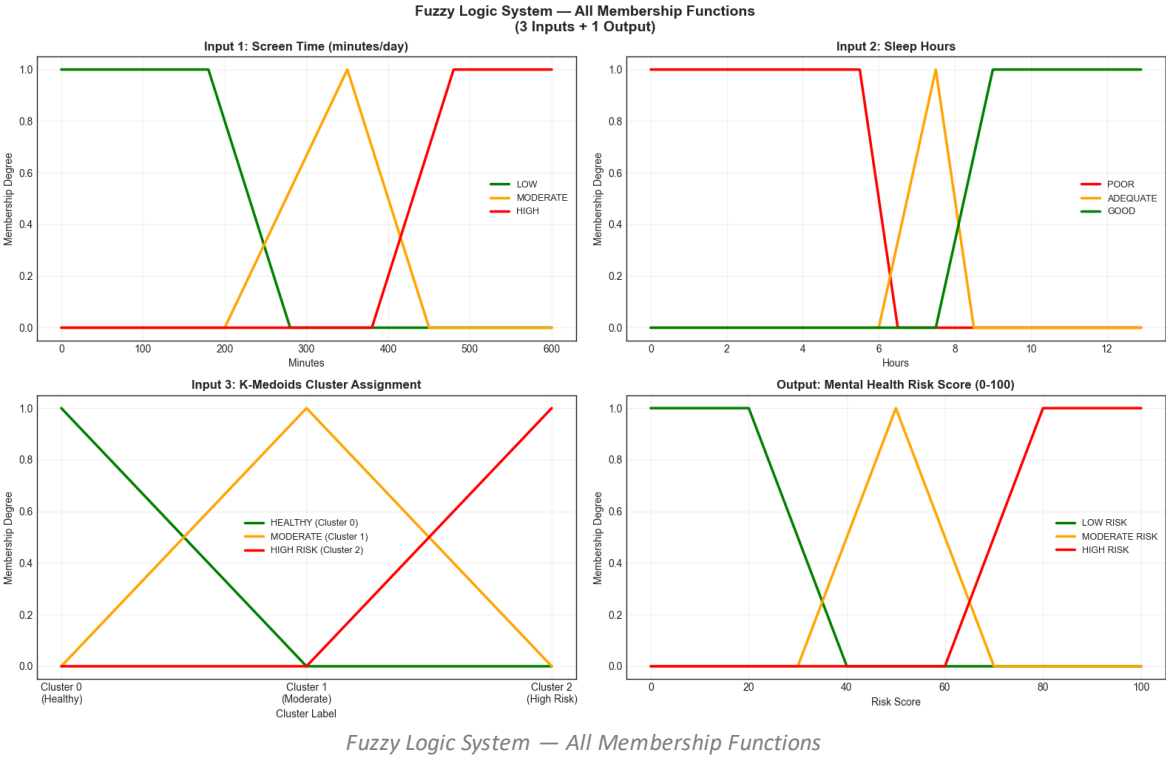
- ❑ LOW: Trapezoidal, covering 0 to 280 minutes (healthy range, under approximately 4.7 hours per day)
- ❑ MODERATE: Triangular, peaking at 350 minutes (approximately 5.8 hours per day)
- ❑ HIGH: Trapezoidal, covering 380 to 600 minutes (clinically concerning range above approximately 6.3 hours per day)

Sleep Hours membership functions:

- ❑ POOR: Trapezoidal, covering 0 to 6.5 hours (below clinical recommendation)
- ❑ ADEQUATE: Triangular, peaking at 7.5 hours (normal adult sleep range)
- ❑ GOOD: Trapezoidal, covering 7.5 to 13 hours (protective factor for mental health)

Risk Score output membership functions:

- LOW RISK: Trapezoidal, covering 0 to 40
- MODERATE RISK: Triangular, peaking at 50
- HIGH RISK: Trapezoidal, covering 60 to 100



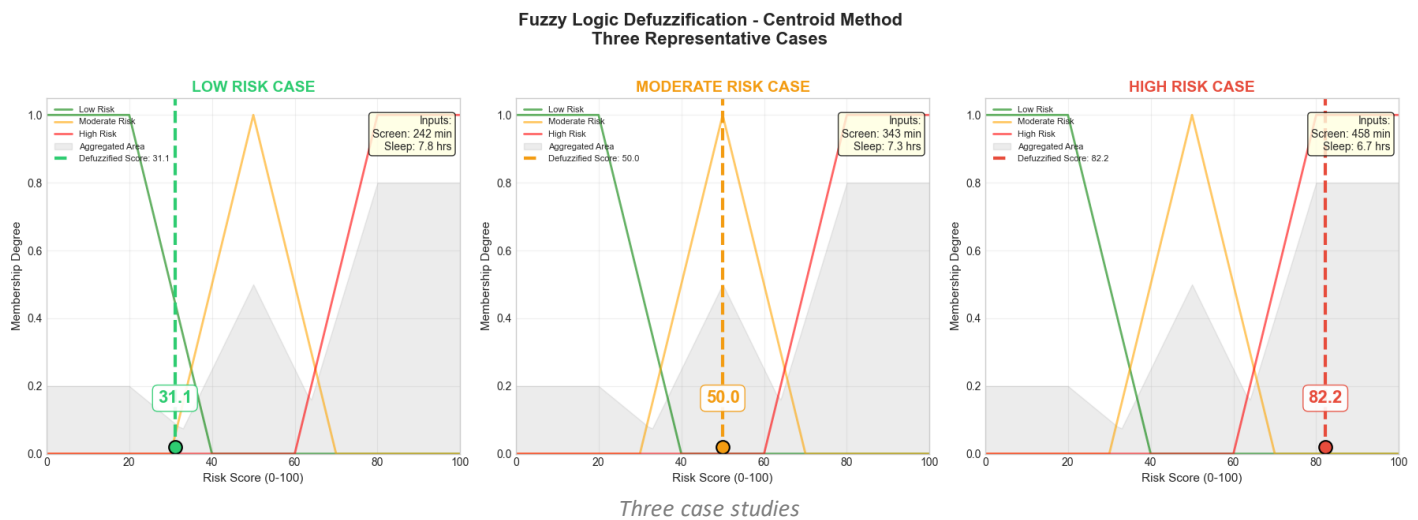
3 Rule Base

Rule	Screen Time	Sleep Hours	Risk Output
R1	LOW	GOOD	LOW RISK
R2	LOW	ADEQUATE	LOW RISK
R3	LOW	POOR	MODERATE RISK
R4	MODERATE	GOOD	LOW RISK
R5	MODERATE	ADEQUATE	MODERATE RISK
R6	MODERATE	POOR	HIGH RISK
R7	HIGH	GOOD	MODERATE RISK
R8	HIGH	ADEQUATE	HIGH RISK
R9	HIGH	POOR	HIGH RISK

The rule base encodes the clinical intuition that screen time and sleep have interactive rather than independent effects on mental health. Sleep functions as a protective factor: a high-screen-time individual who maintains good sleep quality is classified as Moderate Risk rather than High Risk (Rule 7), whereas the same screen time combined with poor sleep yields High Risk (Rule 9). Conversely, low screen time with poor sleep produces only Moderate Risk (Rule 3), reflecting the empirical finding that sleep deprivation alone is less harmful than the combination of excessive digital engagement and sleep deprivation.

4 Defuzzification and Validation

The centroid defuzzification method computes the center of gravity of the aggregated output membership function, producing a precise numeric risk score. The system was validated against all 5,000 records in the dataset, confirming that fuzzy risk scores align correctly with K-Medoids cluster profiles: Cluster 1 (Healthy) produced the lowest mean fuzzy scores, Cluster 0 (Moderate) produced intermediate scores, and Cluster 2 (High Risk) produced the highest mean fuzzy scores.

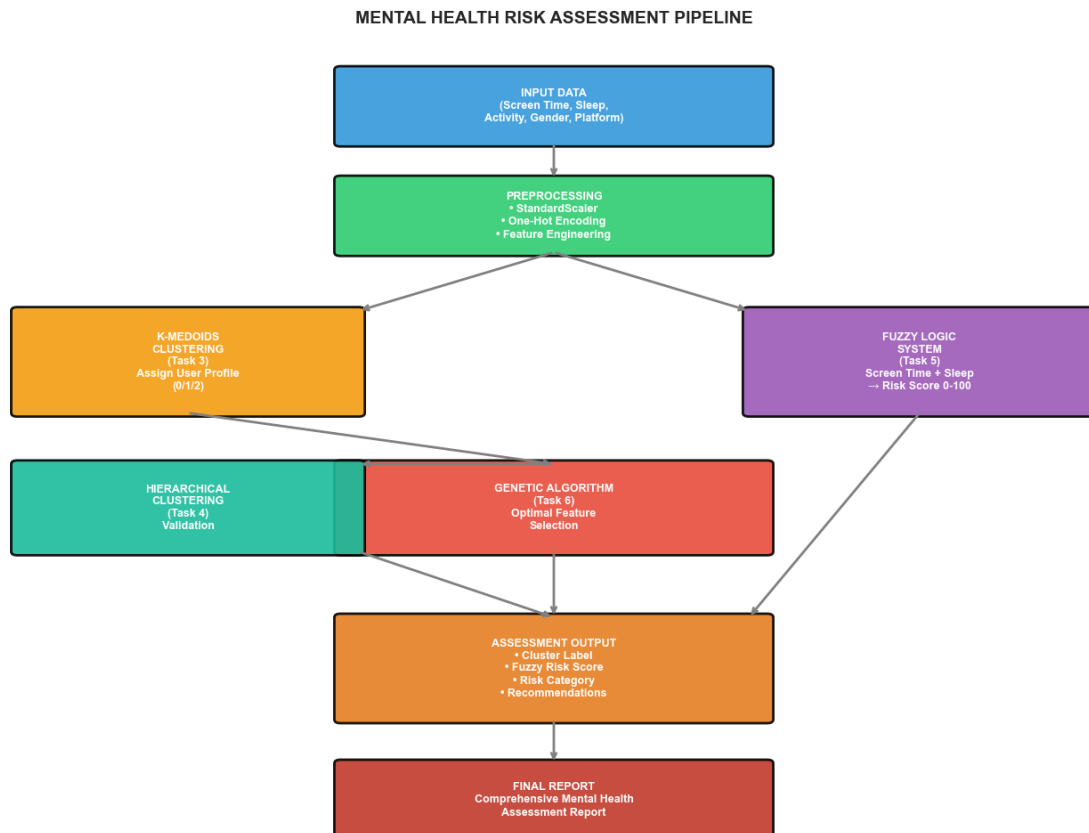


Task 7 — System Implementation Pipeline

The final stage of the project integrates all components into a unified assessment pipeline implemented as a Python class — `MentalHealthPipeline`. The pipeline accepts a new individual's raw data record together with the trained K-Medoids model, the fitted `StandardScaler`, the Fuzzy Logic simulator, and the GA-selected feature list, and produces a complete mental health assessment report.

1 Pipeline Stages

- Stage 1 — Input Construction: The raw input record is converted into the full feature vector matching the structure of the training data, including one-hot encoded gender and platform columns.
- Stage 2 — Preprocessing: Numeric features are scaled using the fitted `StandardScaler`, ensuring that new records are processed identically to the training data.
- Stage 3 — Cluster Assignment: The scaled feature vector is filtered to the GA-selected subset and passed to the K-Medoids model, which assigns the individual to one of the three clusters.
- Stage 4 — Fuzzy Risk Scoring: Raw screen time and sleep hours values are passed to the Fuzzy Logic simulator, which computes a continuous risk score from 0 to 100 via centroid defuzzification.
- Stage 5 — Report Generation: The cluster label, risk score, risk category, and personalized recommendations are assembled into a structured assessment report.



2 Test Case Results

Profile	Screen Time	Sleep Hours	Anxiety	Risk Score	Category	Recommendations
Healthy	160 min	8.5 hrs	1 / 4	16.9 / 100	Low Risk	Maintain healthy lifestyle
Moderate	370 min	7.0 hrs	3 / 4	50.0 / 100	Moderate Risk	Seek support, reduce screen time
High Risk	510 min	5.5 hrs	4 / 4	84.4 / 100	High Risk	Seek support, reduce screen time, improve sleep

The test cases demonstrate the system's ability to differentiate risk profiles across the full spectrum. The healthy profile, characterized by low screen time and adequate sleep, receives a risk score of 16.9 and a single maintenance recommendation. The moderate profile receives a score of 50.0, correctly triggering both a professional support recommendation and a screen time reduction suggestion. The high-risk profile, with the highest screen time and lowest sleep in the test set, receives a score of 84.4 and a comprehensive set of intervention recommendations including sleep hygiene improvement.

3 Output Interpretation Guide

- Risk Score 0 to 35: Low Risk — individual behavioural patterns are broadly consistent with good mental health. Recommendation: maintain current habits.
- Risk Score 35 to 65: Moderate Risk — behavioural indicators suggest emerging risk. Recommendation: consider professional consultation and behavioural adjustments.
- Risk Score 65 to 100: High Risk — behavioural profile indicates significant risk of adverse mental health outcomes. Recommendation: seek professional support, reduce screen time, and improve sleep hygiene.

Conclusion

1 Summary of Findings

This study successfully applied a multi-stage data mining pipeline to identify behavioral criteria associated with mental health risk in a population of 5,000 social media users. The Genetic Algorithm identified nine features as optimal for clustering, including age, screen time, sleep hours, physical activity, mood level, sleep deficit, activity per screen hour, and platform indicators for Instagram and WhatsApp. These features were meaningfully interpretable and consistent with established findings in digital health research.

K-Medoids and Hierarchical clustering both converged on three distinct behavioral profiles with approximately 92% agreement. The three clusters — Moderate Stress, Healthy/Resilient, and High Risk — differed substantially along dimensions of screen time, sleep quality, physical activity, age, and psychological indicators. The High Risk cluster, comprising 42.8% of the dataset, was characterized by the youngest age profile and the heaviest digital engagement, highlighting younger users as the most vulnerable group.

The Fuzzy Logic Inference System produced risk scores that correctly ordered the clusters from lowest to highest risk, demonstrating that screen time and sleep hours alone are sufficient to produce a meaningful and validated risk stratification when combined through a domain-informed rule base.

2 Research and Practical Value

The findings carry several implications for research and practice. For public health practitioners, the identification of young users with heavy screen engagement and poor sleep as the highest-risk group suggests that targeted digital well-being interventions should prioritize this demographic. For platform designers, the consistent association of content-consumption platforms such as TikTok and Instagram with higher stress — contrasted with the better mental health profiles of communication-focused platforms such as WhatsApp — provides evidence relevant to platform design ethics and feature prioritization.

For the data mining research community, this project demonstrates the value of combining unsupervised clustering with evolutionary feature selection and fuzzy inference as a principled framework for behavioral health analytics. The pipeline is generalizable: with an appropriate dataset, the same architecture could be applied to other behavioral health domains such as substance use, sleep disorders, or physical activity monitoring.

3 Limitations

- The dataset is cross-sectional and self-reported, meaning causal relationships between social media usage and mental health cannot be established from this analysis alone.

- The original mental_state labels may not fully reflect clinical diagnostic criteria, limiting the interpretability of comparisons between clustering results and original labels.
- The Fuzzy Logic rule base relies on domain knowledge and design choices that introduce subjectivity into the risk scoring process.